Solar, Wind and Biomass Energy Efficiency Analysis Using Python

Project proposal | Exploratory and Statistical Analysis of Non-Conventional Energy Sources in India

1. Project Overview

This project explores a non-conventional energy optimizer dataset to compare solar, wind, and biomass energy scenarios across seven Indian states. Using Python, the study will examine environmental conditions, installation and maintenance costs, output efficiency, and optimized power output. Tamil Nadu will receive a dedicated state-level comparison. The focus is descriptive and exploratory; predictive machine learning is optional, not required.

2. Data Source and Validation

Dataset file: **non_conventional_energy_optimizer_100k.csv** (user-supplied CSV). Original publisher URL and whether the records are synthetic should be documented and confirmed before final submission. These records should not be described as measured government electricity generation.

Property	Verified from CSV
Total rows	100,000
Original columns	11
File format	Single CSV
Missing values	0
Fully duplicated rows	0
Tamil Nadu records	14,417
Energy categories	wind, solar, biomass
States / locations	7

3. Problem Statement

Energy technologies operate under different environmental and cost conditions. This project asks: How do solar, wind, and biomass scenarios differ in modeled output efficiency, optimized output, and estimated costs across Indian states, particularly Tamil Nadu? Observed patterns will be treated as associations within this dataset, not proof of real-world performance or causality.

4. Project Objectives

• Validate the CSV structure, data types, value ranges, missingness, and duplicate records.

• Compare counts and distributions of solar, wind, and biomass records.

• Explore temperature, wind speed, solar radiation, humidity, and altitude.

• Compare installation and yearly maintenance costs by source type and state.

• Investigate output efficiency and optimized output in kilowatts.

• Study the Tamil Nadu subset against the remaining states.

• Compute descriptive statistics, grouped summaries, and correlation coefficients.

• Prepare at least 15 interpretable Python visualizations and evidence-based findings.

5. Attribute (Column) Details

Column	Meaning
source_type	Solar, wind, or biomass energy category
location	Indian state name
avg_temperature_C	Average temperature in degrees Celsius
wind_speed_mps	Wind speed in meters per second
solar_radiation_kWh_m2	Solar radiation in kWh per square meter
humidity_percent	Relative humidity percentage
altitude_m	Altitude in meters
installation_cost_usd	Estimated installation cost in USD
maintenance_cost_usd_year	Estimated annual maintenance cost in USD
output_efficiency_percent	Reported output efficiency percentage
optimized_output_kw	Modeled optimized output in kilowatts

6. Tools and Technologies

Tool	Purpose
Python / Google Colab	Notebook-based analysis
Pandas	CSV loading, filtering, data validation, aggregation
NumPy	Numerical summaries and feature creation
Matplotlib and Seaborn	Charts and statistical visualization
GitHub	README and project deliverables

7. Data Preprocessing and Feature Engineering

Data cleaning plan: Load with pandas; inspect shape, info, data types and ranges; check missing cells, duplicated rows, unexpected categories, impossible numerical values and outliers. The supplied CSV currently has **zero NULL cells and zero fully duplicated rows**; the report will not falsely claim missing-value imputation or duplicate removal was performed. Do not artificially introduce missing values simply to demonstrate cleaning.

Possible derived features: installation cost per optimized kW (where output is positive), annual maintenance-to-installation cost ratio, temperature bands, wind-speed bands, solar-radiation bands, efficiency bands, and a Tamil Nadu versus other states flag. These are analytical calculations, not new source observations.

8. Exploratory Data Analysis (EDA)

• Record count by source type and by state; Tamil Nadu subset size.

• Summary statistics (mean, median, minimum, maximum, standard deviation, quartiles).

• Distribution of output efficiency, optimized output, installation and maintenance costs.

• Source-type and state-wise grouped means and medians.

• Numerical correlations and unusual-value checks.

• Data-quality audit: missing values, duplicates, category spelling and range validity.

9. Statistical Analysis

Method	Application
Mean / median / mode	Typical cost, efficiency, output and category
Min / max / range	Environmental and financial limits
Standard deviation / variance	Variability of output and cost
Quartiles / IQR	Distribution spread and outlier review
GroupBy / pivot tables	Energy type and state comparisons
Pearson / Spearman correlation	Relationships among numeric fields

10. Data Visualization Plan (15 Charts)

A. Univariate (5 charts)

• 1. Energy source distribution - count plot.

• 2. State-wise record counts - horizontal bar chart.

• 3. Output efficiency distribution - histogram.

• 4. Optimized output distribution - histogram.

• 5. Installation cost distribution - histogram.

B. Bivariate (5 charts)

• 6. Average output efficiency by energy type - bar chart.

• 7. Median optimized output by energy type - bar chart.

• 8. Installation cost by energy type - box plot.

• 9. Wind speed versus optimized output - scatter plot.

• 10. Solar radiation versus optimized output - scatter plot.

C. Multivariate (5 charts)

• 11. Numerical feature correlation - heatmap.

• 12. Energy type and state versus mean efficiency - grouped bar chart.

• 13. Installation cost versus optimized output by source type - scatter plot.

• 14. Temperature versus efficiency by source type - scatter plot.

• 15. Tamil Nadu versus other states: output by energy type - grouped bar chart.

Chart choice will be adjusted if the underlying relationships are not meaningful; for example, wind-speed comparisons can be limited to wind records and solar-radiation comparisons to solar records.

11. Planned Findings and Interpretation

The completed project will report at least five findings grounded in computed results, covering energy-source differences, cost and output distributions, environmental associations, state-wise patterns, and Tamil Nadu comparisons. No numerical conclusion or technology ranking is claimed before completing EDA. Correlation will not be interpreted as causation.

12. Types of Analysis

Type	Planned work
Descriptive	Counts, averages, spread, distributions, state-wise summaries
Diagnostic / exploratory	Investigate patterns between environmental, cost and output fields
Predictive	Not required; optional future modeling
Prescriptive	Data-informed opportunities for further evaluation, with dataset limitations

13. Future Enhancements

• Validate conclusions using real-world datasets from government or grid operators.

• Add dates, plant capacity and actual electricity generation when available.

• Normalize costs using a common reference year and currency.

• Extend to renewable energy forecasts or a simple efficiency model after validation.

• Create a lightweight dashboard for interactive state comparisons.

14. Conclusion

This proposed Python project demonstrates dataset validation, descriptive statistics, exploratory data analysis, feature engineering and 15 planned visualizations across solar, wind and biomass scenarios. Its main value is a structured comparison of modeled energy efficiency, output and costs, including a Tamil Nadu-focused analysis. The report will clearly distinguish dataset-derived observations from real-world energy performance and will be finalized with actual figures after the analysis is completed.

Approval note: Dataset is large (100,000 rows), fully populated (0 missing cells), and should be treated as a scenario/optimizer dataset until the original provenance is confirmed. Instructor approval is recommended before full implementation.